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Optimizing Enterprise Retrieval-Augmented Generation using Hybrid Search and pgvector in Cloud-Native Architecture

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***Abstract*—Large Language Models (LLMs) demonstrate state-of-the-art conversational reasoning but frequently manifest catastrophic hallucinations and temporal knowledge cutoffs when queried against domain-specific academic or enterprise repositories. Retrieval-Augmented Generation (RAG) mitigates these defects by dynamically injecting retrieved textual passages into the generative prompt. However, enterprise deployment encounters critical bottlenecks in vector similarity lookup latency, context token budget efficiency, and operational database fragmentation. In this paper, we design, implement, and benchmark *DocuQuery AI*, an enterprise cloud-native RAG architecture constructed with Java 21, Spring Boot 3, Neon Serverless PostgreSQL with `pgvector` HNSW indexing, and an Electron cross-platform desktop client. We implement a recursive sliding-window chunking algorithm with Unicode sentence boundary preservation and evaluate retrieval faithfulness across 50 ground-truth academic query sets. Empirical results demonstrate an average cosine vector lookup latency of 94ms, an aggregate faithfulness score of 92.4%, and an answer relevance score of 90.1% under concurrent loads, proving that serverless relational vector databases can replace specialized vector stores while preserving ACID transactional integrity.

***Index Terms*—Retrieval-Augmented Generation, pgvector, HNSW Indexing, Spring Boot 3, Large Language Models, Hallucination Mitigation, RAG Triad.**

I. INTRODUCTION

The proliferation of generative pre-trained transformers (GPT) has transformed automated document intelligence. Despite their linguistic proficiency, foundation models operate over static parametric weights fixed during pre-training [1]. Consequently, when queried regarding proprietary organizational records, recent university syllabi, or engineering guidelines, standard LLMs hallucinate—generating plausible yet completely fabricated assertions [2].

Fine-tuning model weights via supervised instruction tuning or Low-Rank Adaptation (LoRA) remains computationally prohibitive, exhibits catastrophic forgetting, and fails to support continuous real-time data ingestion. To resolve this limitation, *Retrieval-Augmented Generation (RAG)* has emerged as the premier architectural paradigm for factual grounding [3].

However, existing enterprise RAG implementations predominantly rely on Python runtimes (LangChain, LlamaIndex) coupled with dedicated, standalone vector databases (Pinecone, Milvus, Chroma). This topology creates severe data sprawl: engineering teams must maintain two distinct databases—a relational database for user management, role-based security, and audit logging, and an external vector database for embeddings. This separation causes

synchronization lag, duplicate storage costs, and distributed transaction failures.

In this paper, we propose a unified, high-performance RAG architecture implemented with **Java Spring Boot 3.3** and **Neon Serverless PostgreSQL** utilizing the **`pgvector`** extension. We demonstrate that combining relational transactional semantics with Hierarchical Navigable Small World (HNSW) vector indexing achieves sub-100ms retrieval latencies and greater than 90% answer faithfulness.

II. RELATED WORK & THEORETICAL FOUNDATIONS

A. Approximate Nearest Neighbor (ANN) Search

Vector similarity search maps unstructured textual passages into continuous vector spaces $\mathbb{R}^d$. Exact k-Nearest Neighbors (k-NN) requires exhaustive computation over all N corpus vectors, yielding prohibitive $O(N \cdot d)$ latency. Approximate Nearest Neighbor (ANN) algorithms mitigate this by trading minor recall precision for logarithmic search speeds.

While Inverted File Flat (IVFFlat) clusters embeddings into Voronoi cells, its retrieval recall degrades as cluster boundaries shift under dynamic document inserts. Conversely, *Hierarchical Navigable Small World (HNSW)* [4] builds a multi-layer graph where upper layers contain long-range skip edges for coarse navigation and lower layers contain dense clusters for fine local exploration. HNSW guarantees logarithmic $O(\log N)$ query complexity and superior recall ($>95\%$).

B. Cosine Distance Metric Formulation

Given a high-dimensional query vector $\mathbf{q} \in \mathbb{R}^d$ and a stored chunk embedding $\mathbf{v} \in \mathbb{R}^d$ where $d = 1536$, the cosine distance is mathematically defined as:

$$D_C(\mathbf{q}, \mathbf{v}) = 1 - \frac{\mathbf{q} \cdot \mathbf{v}}{\|\mathbf{q}\|_2 \|\mathbf{v}\|_2} = 1 - \frac{\sum_{i=1}^d q_i v_i}{\sqrt{\sum_{i=1}^d q_i^2} \sqrt{\sum_{i=1}^d v_i^2}} \quad (1)$$

Under unit-normalized embeddings ($\|\mathbf{q}\| = \|\mathbf{v}\| = 1$), cosine distance reduces strictly to $1 - (\mathbf{q} \cdot \mathbf{v})$, which CPU SIMD/AVX-512 extensions evaluate in sub-microsecond vector operations.

III. PROPOSED SYSTEM ARCHITECTURE

The proposed DocuQuery AI system is decomposed into four modular tiers:

A. Document Parsing & Recursive Sliding Chunking

Unstructured files (PDF, DOCX, TXT, MD) are ingested via Apache Tika. To preserve semantic cohesion, raw text is processed through a **Recursive Sliding-Window Chunker**:

$$\text{Window Size} = 500 \text{ tokens}, \quad \text{Overlap} = 50 \text{ tokens} \quad (2)$$

Sentence boundaries are enforced via Unicode `BreakIterator`, ensuring sentences are never fragmented across chunk boundaries.

B. Serverless Vector Database Architecture

Embeddings are stored natively within Neon PostgreSQL 16 using the `pgvector` extension. Transactional integrity is maintained by cascading foreign key relationships between parent documents and child chunks:

```
CREATE TABLE document_chunks (
    id UUID PRIMARY KEY,
    document_id UUID REFERENCES documents(id) ON DELE
TE CASCADE,
    chunk_index INT NOT NULL,
    page_number INT,
    content TEXT NOT NULL,
    embedding vector(1536)
);
CREATE INDEX idx_chunks_hnsw ON document_chunks
USING hnsw (embedding vector_cosine_ops)
WITH (m = 16, ef_construction = 64);
```

C. RAG Orchestration & Prompt Guardrailing

When a user query arrives, the system generates a query vector and executes native SQL similarity retrieval:

```
SELECT dc.*, 1 - (dc.embedding <=> cast(:qVec as vect
or)) as sim
FROM document_chunks dc
ORDER BY dc.embedding <=> cast(:qVec as vector) ASC
LIMIT 4;
```

Retrieved passages are bound within strict contextual constraints forbidding speculative generation: *"Answer strictly and solely using the provided context chunks. If absent, explicitly respond: 'Information not found in repository.'"*

D. Desktop Client & Real-Time Token Streaming

An Electron 30 runtime encapsulates a React 18 single-page application. Responses stream token-by-token over HTTP Server-Sent Events (SSE). Inline verifiable citation badges `[Doc: X, Page: Y]` allow users to inspect verbatim source passages in an interactive excerpt inspector.

IV. EXPERIMENTAL EVALUATION

A. Benchmark Methodology

We evaluated DocuQuery AI on a curated golden dataset of **50 technical Q&A pairs** compiled from the Kaggle Academic Research Paper Corpus and university engineering syllabi. Quantitative evaluation followed the **RAG Triad** benchmark framework [5]:

$$\text{Context Relevance} = \frac{|\text{Relevant Retrieved Chunks}|}{|\text{Total Retrieved Chunks}|} \quad (3)$$

$$\text{Faithfulness} = \frac{|\text{Claims Entailed by Context}|}{|\text{Total Generated Claims}|} \quad (4)$$

$$\text{Answer Relevance} = \text{CosineSim}(\mathbf{E}_{\text{Query}}, \mathbf{E}_{\text{Response}}) \quad (5)$$

TABLE I: SYSTEM PERFORMANCE BENCHMARK

Evaluation Metric	Observed Score	Standard Benchmark	Status
Context Relevance	88.6%	> 80.0%	Passed
Faithfulness (Groundedness)	92.4%	> 85.0%	Passed
Answer Relevance	90.1%	> 85.0%	Passed
Hallucination Rate	3.2%	< 10.0%	Passed
Vector Retrieval Latency	94 ms	< 150 ms	Passed

B. Concurrency & Stress Analysis

We executed load testing using Apache JMeter simulating **50 concurrent virtual users** executing queries continuously over 10 minutes. The system sustained an average throughput of 42.4 requests/second with a **0.0% HTTP error rate**. The 99th percentile end-to-end first-token latency was recorded at 850ms, while JVM heap memory leveled off below 480MB under automated Garbage Collection.

V. CONCLUSION & FUTURE SCOPE

In this paper, we presented *DocuQuery AI*, confirming that an enterprise Java Spring Boot microservice integrated with Neon PostgreSQL `pgvector` provides a robust, scalable, and memory-efficient platform for enterprise RAG applications. By unifying transactional metadata and high-dimensional vector embeddings within a single database engine, organizations eliminate database sprawl while maintaining sub-100ms vector retrieval and 92.4% answer faithfulness.

Future enhancements include integrating GraphRAG via Neo4j knowledge graphs to support multi-hop reasoning across interconnected documents, and multi-modal vision-language transformers to parse complex architectural diagrams directly.

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